Date: 10/10/2026

CO1 ASSESSMENT TOOL 1 - APPLICATION BASED QUESTION

DSA0308-NLP

Reg no: 192425342 Name: Raihan Hussain

1. Resume Information Extraction Using Regular Expression

Explanation:

The resume information extraction system is developed using Python and regular expression to automatically extract candidate details.

1. **Extract Candidate Name:** Identify and extract the candidate's name from the resume text.

2. **Identify Email Address:** Uses regular expressions to find the candidate's email address.

3. **Extract Mobile Number:** Detects and extracts the candidate's mobile number from the resume.

4. **Detect Technical Skills:** Identifies technical skills such as Python, Java, SQL, Machine Learning and NLP.

5. **Extract Years of Experience:** Finds the candidate's work experience in years from the resume.

6. **Generate Structured Summary:** Organizes the extracted details, including name, email, mobile number, skills, and experience, into a clear candidate profile.

7. **Check Eligibility Criteria:** Shortlists candidates who have at least 2 years of experience and possess Python skills. Candidates who do not satisfy both conditions are marked as not eligible.

Result:

The system extracts candidate information automatically, generates a structured profile and identifies eligible candidates, reducing manual work and saving time for the HR department.

2. Product Search System Using Regular Expressions

Explanation:

The product search system is developed using Python and Regular Expressions to help customers find products easily in an e-commerce application.

1. **Exact Keyword Search:** Searches for a product that exactly matches the given keyword.

2. **Prefix-Based Search:** Finds products whose names start with the specified keyword.

3. **Suffix-Based Search:** Finds products whose names end with the specified keyword.

4. **Partial Keyword Search:** Finds products containing the given keyword anywhere in their names.

5. **Case-Insensitive Search:** Searches for products without considering uppercase or lowercase letters.

6. **Display Matching Products:** Displays all product names that match the given search conditions.

7. **Generate Search Report:** Counts and displays the total number of matching products for each type of search.

Result:

The system performs flexible product searches using Regular Expressions, displays matching product names, and generates a report showing the total number of matching products for each search.

3. University Registration System Using Regular Expressions

Explanation:

1. A university registration system is developed to verify student information before allowing students to enroll in courses.

2. The system collects important student details, including the register number, institutional email address, course code, semester information, and mobile number.

3. **Register Number Validation:** The system checks whether the student's register number follows the format specified by the university.

4. **Institutional Email Validation:** It verifies whether the student's email address is correctly formatted and belongs to the university's official email domain.

5. **Course Code Validation:** The system checks whether the entered course code contains the required letters and numbers according to the university's rules.

6. **Semester Validation:** It verifies whether the entered semester is within the permitted range, such as semesters 1 to 8.

7. **Mobile Number Validation:** The system checks whether the mobile number contains the required number of digits and follows the specified format.

8. **Validation Messages:** After checking each field, the system displays an appropriate message indicating whether the entered information is valid or invalid.

9. **Final Registration Status:** The system examines the validation results of all required fields. If every field is valid, the registration is marked as successful. If any field is invalid, the registration is marked as unsuccessful.

10. **Report Generation:** Finally, the system displays a registration report containing the validation status of each field and the overall registration result.

Result:

The university registration validation system verifies student details using Regular Expressions, identifies invalid information, displays appropriate validation messages, and determines whether the registration is successful.